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First Remote Operations of an Unmanned Platform in the Middle East with an Autonomous Inspection Robot

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Abstract

The primary objective of this paper is to demonstrate the ability and potential value of conducting continuous inspection activities in unmanned wellhead production platforms through integration of robotics and Artificial Intelligence (AI) in Oil & Gas hazardous environments to significantly reduce personnel exposure, particularly in offshore and remote locations where access is costly.

TotalEnergies initiated an extensive qualification program in 2020 to operate assets remotely with the assistance of robotics. To date, this program has included eight pilot tests. The previous pilot in Angola, demonstrated technology's scalability for inspecting different types of equipment in a remote, congested FPSO despite the reliability challenges. The latest successful test, completed in 2025 on an unmanned wellhead production platform in Qatar, involved one robot in operation for eight months, supervised from a Central Control Room or main office in Doha. The methodology focused on continuous inspection activities to validate robotics effectiveness for routine surveillance and inspection tasks.

The Qatar pilot demonstrated significant advancements in remote operations. Over the testing period, the robot completed autonomous operation cycles of up to 14 days without human intervention, achieving an average Mean Time Between Intervention 2.5 higher than former test on the FPSO, only with human supervision located dozens of kilometers away from the site. The robot inspected more than 500 elements in a single day and reliability improved by 27% from previous pilot, despite the harsh environment. Night shift inspection and stair climbing on two different platform decks were also successfully tested. Intensive use of data analytics enabled efficient interpretation of local information, which would otherwise require personnel mobilization. The Information Technology architecture chosen proved to be scalable and adaptable to the industrial network constraints.

Some areas for improvement were identified, reliable and precise robot localization is essential for building trust in autonomous inspections, robot utilities (charging station, Wi-Fi, nitrogen, and electricity supply) require the same level of attention and performance to ensure successful operations. While robotic solutions remain costly to deploy, the findings suggest that continued development and implementation of robotic and AI systems can lead to safer, more efficient, and cost-effective operations in the Oil & Gas industry.

The integration of robotics and artificial intelligence in routine operations offers new opportunities to enhance safety and operational performance in the energy industry. This paper details the main achievements and challenges encountered during the first long-duration pilot test conducted in the Middle East, on an unmanned wellhead platform.

Introduction

Offshore oil and gas operations are inherently complex, facing persistent challenges in safety, cost-efficiency, and environmental impact. Facilities located in remote regions, particularly in the Middle East, must contend with harsh conditions and logistical constraints. Key issues include exposure to hazardous environments, such as working at heights or handling volatile hydrocarbons, and the risks associated with marine transport during adverse weather. Severe conditions can halt access for days, disrupting inspections and maintenance. Additionally, aging infrastructure and declining production make it increasingly difficult to justify the cost of maintaining full crews to operate remote offshore platforms, pressing operators to find more efficient ways to run late-life assets. Frequent transport of personnel and materials also contributes significantly to greenhouse gas emissions, conflicting with the industry's sustainability goals.

Towards Remote Operations, Robotics-Driven Solution

To address these issues, TotalEnergies is developing a new operating model: the Normally Unattended Facility (NUF). This approach leverages autonomous mobile systems to perform routine tasks on offshore platforms, reducing the exposure of personnel to hazardous environments. Robots and drones can conduct inspections, collect data, and assist in emergency checks, all under remote supervision from onshore teams. This operating philosophy directly tackles the industry challenges by decoupling on-site work from the presence of personnel on the facility.

Recent advances in autonomous navigation, telecommunications, high-performance computing, and artificial intelligence have enabled the development of robots capable of executing repetitive, low-value tasks with minimal human intervention. These technologies support the progressive implementation of NUFs, offering a safer, more efficient, and environmentally responsible alternative to traditional offshore operations.

Proven Feasibility and Outlook

Since 2020, the company has conducted a series of pilot programs to validate this model. The most recent and ambitious trial took place in Qatar between 2024 and 2025, where a mobile inspection robot operated continuously for eight months on an unmanned wellhead platform DP4 belonging to Al-Khalij field development. Supervised remotely from Halul Island and Doha, the robot performed operator rounds, conducted surveys, monitored gas presence, and collected local operational data for onshore teams. It completed hundreds of inspections per day, including stair climbing and night operations, demonstrating resilience in harsh offshore conditions and the ability to execute autonomous inspection without human intervention for up to two weeks.

The supporting digital infrastructure, including wireless networks and remote analytics platforms, proved robust and secure throughout the trial. The Qatar pilot confirmed that integrating robotics and AI can enable continuous monitoring with zero personnel on site, directly addressing safety and access limitations. While the technology is still evolving and initial deployments require significant investment, each iteration improves reliability and builds trust. Challenges such as navigation in complex environments and enhancement of robot ancillary systems are actively being addressed.

In summary, autonomous inspection robots are redefining offshore operations. Their proven viability marks a pivotal step toward making NUFs a practical solution in the short to medium term for oil and gas site operations. The following sections of this paper will explore the specific challenges addressed, the

robotic systems deployed, and the outcomes of the Middle East pilot, illustrating how remote operations were safely and effectively realized.

Statement of Theory and Definitions

Qatar Specific Context

The pilot project in Qatar represented the first long-term deployment of autonomous robots on an offshore unmanned wellhead platform DP4 operated with no routine crew on board. Instead, control is maintained remotely, and maintenance supported locally by the operations team located in Halul Island. The region can experience up to 100 "non-sailing" days per year when rough seas prevent crew transport. The DP4 platform itself spans four deck levels (each about 30 m by 30 m), requiring the robots to navigate stairs and cluttered areas, sometimes even operating alongside visiting personnel (co-activity) in a harsh, marine climate of high heat and humidity.

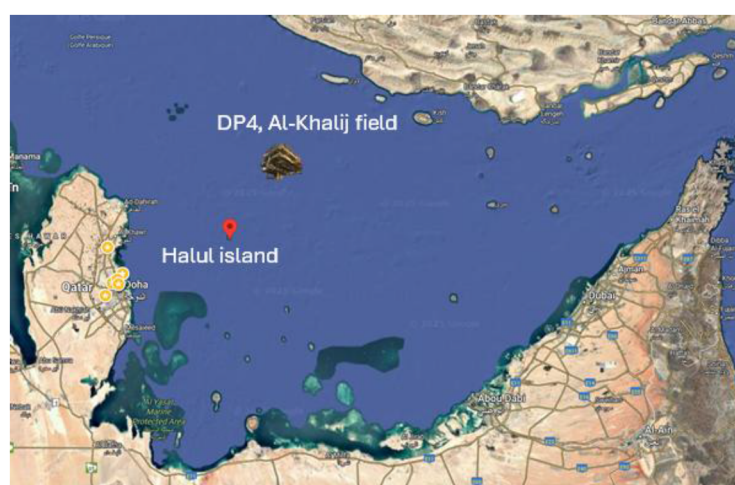


Figure 1—Robotics pilot location

To execute the pilot, TotalEnergies deployed two ATEX-certified robots along with two prototype charging stations on the platform. At any given time, one robot was active and the other on standby, ready to take over if needed, ensuring continuous operation. The pilot was initially structured as a 6-month R&D program on the unmanned platform. The commissioning phase was planned for two weeks in October 2024 to set up and test the system, followed by a six-month execution phase (November 2024 – April 2025). Finally, a decommissioning phase was planned in May 2025 to conclude the trial and remove the equipment.

Pilot Objectives

The project had defined objectives to ensure it delivered value and proved the viability of the NUF model in a real-world setting. The objectives served as success criteria, guiding both the operational use of the robots and the evaluation of the supporting systems.

Perform Remote Inspection Rounds. Use the robot to conduct operator-style inspection patrols, continuously assessing site safety and integrity (checking equipment status, detecting anomalies) without putting humans in hazardous environments.

Enable Teleoperation for Special Tasks. Allow engineers to teleoperate the robot in real time for immediate assistance during emergencies and to conduct on-demand surveys aimed at optimizing production (for example, investigating a suspected issue or gathering data to fine-tune operations).

Monitor for Gas Leaks. Leverage the robot’s sensors to continuously monitor the presence of flammable gas. This included running regular automated gas detection surveys and providing support to emergency response teams by measuring gas levels in case of an incident.

Collect Site Data for Onshore Teams. Stream operational data (equipment readings, images, thermal scans, etc.) from the platform to onshore support teams. This data collection would help experts onshore in decision-making and reduce the need for frequent site visits.

In addition, the pilot had R&D oriented objectives to evaluate the progress of the technology:

Demonstrate that an unmanned facility with robots can operate reliably. Concretely, this meant proving the robot hardware’s reliability in harsh offshore conditions and achieving continuous robot operations for at least 7 days or more without human intervention.

Validate Robotics Ecosystem. Evaluate the performance of the complete robotic ecosystem, not just the robots themselves. This involves the Robot Supervision System (RSS) and Remote Analytics System (RAS) software and IT infrastructure. Key metrics included the mission planning and execution success rate (how reliably the system can plan missions, and the robot complete them) and the effectiveness of data acquisition, transmission, and interpretation. In other words, the pilot measured how well the system collected data via the robot, sent it to the onshore systems, and analysed it to produce useful information.

By the end of the pilot, meeting these technical objectives would indicate tangible progress toward routine unmanned operations, and highlight any improvements needed in hardware or software before a broader deployment.

Robotic System Key Indicators

Implementing an autonomous robotic solution offshore requires measuring success in terms of work accomplished, not just robot functionality. One key insight from TotalEnergies’ approach is to avoid focusing solely on the robot gadget itself and instead evaluate the entire system’s ability to perform useful work on-site. For this pilot, the solution comprised several integrated components — the robots and their charging stations, various sensors, the communication network, the on-site and remote IT infrastructure, the Robot Supervision System (RSS) for mission planning and control, and the cloud-based Remote Analytics System (RAS) for data interpretation.

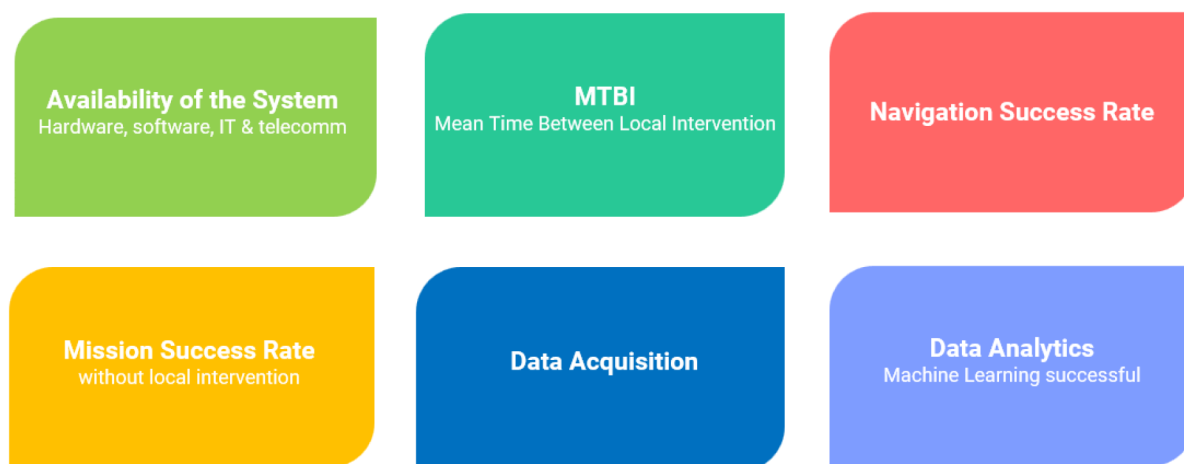


Figure 2—Robotics pilot key parameter indicators

Project Organization

Despite the complexity of the system (robots, devices, networks, and software), a core principle of the project was that day-to-day robot operations should be handled by non-robotics specialists either located

in the Halul Island Central Control Room (CCR) or in Doha main office. Prior to the pilot's start, several training sessions were conducted to prepare the affiliate's robot supervisors (end-users). These trainings familiarized engineers with planning autonomous missions, initiating teleoperated runs from the CCR or the Doha office, performing basic preventive maintenance on the robots, and monitoring system performance.

For more complex tasks, a tiered support model was used. Initial system setup, mission configuration, and any advanced troubleshooting or software adjustments were carried out with the assistance of specialists from TotalEnergies' headquarters, the robot manufacturer, and the software providers.

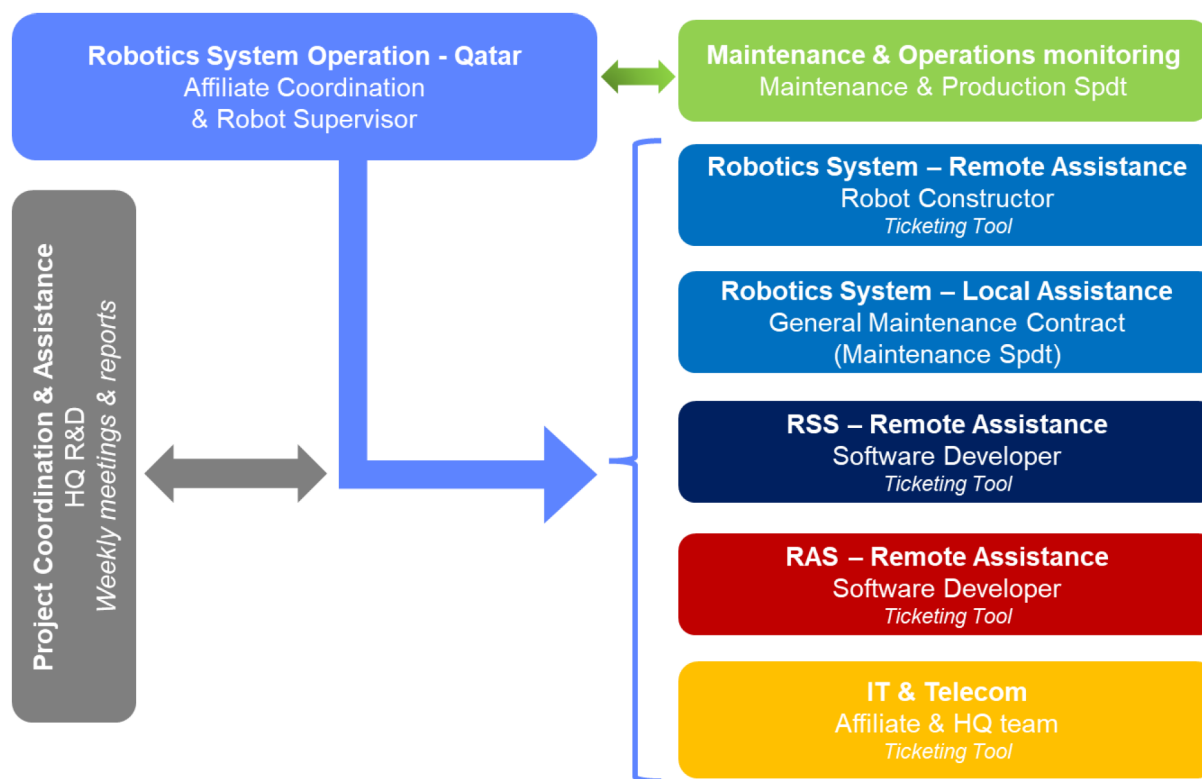


Figure 3—Schematic of the project organization implemented during the pilot

Description and Application of Equipment and Processes

Robotics Hardware

A robust hardware setup was essential to achieve reliable long-term operations. The pilot employed a dual-robot system for redundancy: two identical inspection robots were installed, but only one operated at a time while the second remained on "hot standby" to immediately step in if the active unit encountered a problem. This strategy maximized the chances of the platform always having a functioning robot available to perform tasks, thereby increasing the total work completed over the pilot period. The robot specifications were:

- Taurob Inspector V8
- ATEX zone 1 / IP 67
- Environment temperature: up to 60°C
- Autonomy of the battery: 3h
- Battery charging time: 75min

The robot is comparable to a human in terms of size and mobility: weight 85 kg., height with the arm deployed 1.64 m. and driving speed 2 km/h.

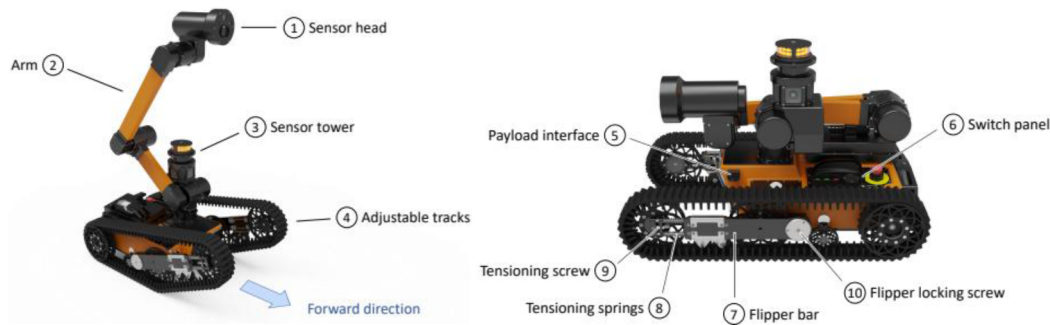


Figure 4—Inspection robot

Charging Stations: To support continuous operation, the trial included two charging stations placed on different levels of the platform: cellar and access deck. So, redundancy would prevent the robotic system to stop carrying out tasks in case one became unavailable but also to allow the robot to charge its battery in the closest unit available.

- Non-ATEX prototype charging station (Wallbox)
- Environment temperature: up to 60°C
- Size: 1335 × 688 × 1410 mm
- Weight: 75kg
- Electrical supply input: 220 V
- Nitrogen feed pressure: 8 barg

The absence of a nitrogen network in DP4 (uncommon in normally inhabited offshore wellhead platforms) required the charging station to be connected and fed with a portable nitrogen bottle.



Figure 5—Charging station fed with N2 bottles

IT Infrastructure

From an IT perspective, the robot deployment was treated as a full-fledged information system implementation, with a multi-layered architecture to handle control, data, and analytics. At the core was the Robotic Supervision System (RSS), this on-premises system managed robot missions. Alongside it ran the vendor's Fleet Management System (FMS), which interfaced directly with the robots for navigation

and docking control. To derive value from the data being collected, the setup included a Remote Analytics System (RSS), a cloud-hosted platform where all inspection data (such as sensor readings, images, videos, sound recordings) was aggregated and analysed using machine learning models. This layered design (robots & sensors → RSS/FMS → RAS) provided modularity and scalability, allowing each component to be improved or expanded independently.

The hardware deployment consisted of two local servers at the Halul site, one running the RSS and the other the FMS. Additionally, operator stations with Human-Machine Interface (HMI) software were installed at both the Halul CCR and in Doha office, enabling operators and engineers at either location to supervise the robot in real time. Because the system spanned across different networks (offshore industrial networks and corporate IT networks), careful attention was paid to cybersecurity and connectivity. The platform's network and the onshore networks were separated by firewalls, and three key firewall zones were defined to segregate the robot system components. A DMZ (Demilitarized Zone) EDGE network was established to securely broker communications between the on-premises systems in Halul Island and TotalEnergies' cloud (where the RAS resided). All data flowing to the cloud was tunnelled through this controlled interface, ensuring that no direct, insecure connections were made from the industrial zone to the internet.

Crucially, the design aligned with TotalEnergies' cybersecurity policies, maintaining a strict separation between Industrial Information Systems (IIS) and Enterprise Information Systems (EIS). The Qatar subsidiary's existing network topology (a star architecture linking sites via secure IPsec tunnels and SD-WAN connections) was leveraged, with the robot system fitting into the established security framework requiring minor exceptions. The entire robotic inspection system was classified under Security Profile 1 (SP1) – indicating a system of moderate criticality with tightly controlled external communications.

Presentation of Data and Results

Key Indicators

During the pilot, KPIs were tracked closely for the last 3 months once the whole robotics system was debugged and stabilized and overall, the results were very encouraging even though some of the goals were not fully met or exceeded.

The availability improved by 27% compared to previous FPSO pilot in Angola, indicating a significant reduction in the downtime of the intended operating time. This was below the ambitious target, because of a few technical issues (discussed later) including a multi-week halt to repair a damaged charging station. On the plus side, when the system was up, it performed 550 missions (860 checkpoints configured) successfully over a period of 67 days, equivalent to 87% of the time. The small gap compared to mission success target were due to solvable problems like the robot encountering navigation or localization issues or unsuccessful docking.

The navigation success 85% is slightly below mission success, implying a few missions required manual intervention specifically for navigation problems. For example, few instances where the robot got confused (lost localization) and could not find the docking station without help; these were infrequent relative to the large number of missions. The team has identified improvements like better visual markers and updates to the navigation algorithm to improve this.

Data acquisition was nearly perfect – the robot reliably gathered what it was instructed to, indicating strong performance of sensors, cameras, and connectivity in transmitting data. Almost 10 000 measures were collected in a period of 67 days, reaching up to 500 checkpoints in a single day with one robot.

Data analytics results suggest that while data was collected, the automated analysis pipeline (machine learning models identifying equipment states, etc.) wasn't fully mature. For example, the robot took many images and readings, but the system did not automatically interpret all of them correctly (false positive anomaly detection), requiring human analysts to review some data. This is an important insight: the robot

can relieve humans from data collection, but making sense of the data is a separate challenge where AI is still catching up.

Finally, the MTBI results on average reflects the fact that in the first couple of months, human interventions were frequent due to debugging and some major outages. However, as the pilot progressed, the periods without intervention improved markedly. In March 2025, after fixes were implemented, the system achieved the full 7 days of continuous autonomy multiple times (five separate weeks) and even a 14-day stretch once. That two-week achievement is a crucial proof-of-concept: it demonstrated that under stable conditions, the platform could run completely unattended for a fortnight. If the early-phase issues (like charging station reliability) were resolved in a future deployment, one could expect the average MTBI to approach or exceed the goal consistently. In short, by the latter half of the execution phase, the pilot was meeting most KPIs, indicating that the concept is viable, even though the early hurdles pulled some averages down.

In summary, the KPI measurements showed significant progress toward a fully remote-operated platform, while also exposing the weak links (notably the docking system and some software elements) that need to be strengthened to reach industrial robustness.

Robot Hardware

In terms of raw hardware reliability, the robots fared impressively well, especially given the conditions. Over the entire pilot, neither robot had a critical mechanical failure. For instance, the tracks on the robots did not need replacement at all, and there was only a single instance of a track derailment, which is remarkable considering months of continuous use. By comparison, in the previous pilot with older robots, track wear and derailments were frequent (tracks lasted ~3 weeks on average in Angola). The improved track design clearly paid off, as did the better dust sealing; DP4's robots handled the abrasive salt/dust exposure without abnormal wear. The new battery system also proved out: each robot's battery consistently delivered over 3 hours of runtime, and the charging cycle was reliable. Battery autonomy was confirmed in real conditions and seen as a major improvement over the prior generation. Stair climbing with a crawler robot type proved initially difficult requiring testing different solutions, from simple coating until installation of anti-slip nosing on the stairs' steps, which in turn allowed safe autonomous navigation through the different platform decks.

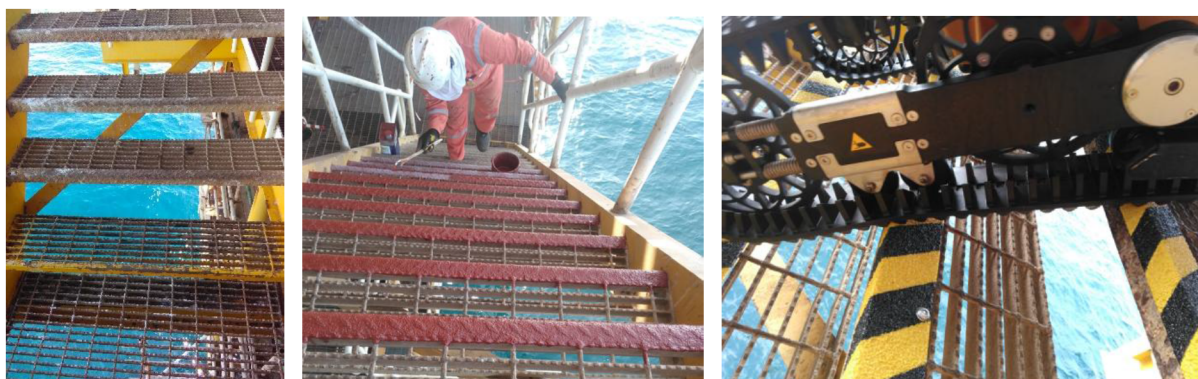


Figure 6—Stairs climbing difficulties: (left) original step noising profile; (center) anti-slip coating; (right) anti-slip nosing

The harsh environment did introduce some maintenance needs: fine dust and salt accumulated on the LiDAR sensor cover, at times partially obscuring the robot's navigation laser. The team found they needed to clean the LiDAR window periodically to maintain optimal navigation. Later, a nano-coating on the glass was added to reduce dirt adhesion. But crucially, heat did not cause any robot shutdowns, neither electronics overheating or seizures due to temperature reported, and no corrosion or water ingress issues despite high

humidity and occasional spray. This gave confidence that the hardware platform is robust enough for Middle East conditions as-is.

For the robotic system to be autonomous long-term, it must reliably dock for recharging. Unfortunately, during the pilot the docking process often failed, especially in the earlier months. The pilot identified multiple factors: an undersized electrical breaker on the charging station power supply, the robot had initially difficulties to see the docking bay in bright sunlight. Dust on the LiDAR also reduced docking precision. The floorplate in front of the dock became slick with offshore grime, causing the robot's tracks to slip. Inaccurate pressure regulator of the nitrogen purging system caused over-pressure in the N₂ supply, damaging components of the charging station. The lesson learned is that the charging station must be made more robust and industrialized – the prototype was not originally designed for months of continuous use in a desert marine environment. A new ATEX-certified, industrial-grade docking station is under development and testing in a safer onshore site before any future offshore deployment.



Figure 7—Charging station configuration: (left) original set-up; (right) without floorplate to avoid robot's tracks slip

The team also observed two instances where the robot mislocalized itself and found the localization algorithm became too permissive, and the robot's software lacked a feature to detect the so-called negative obstacles. As corrective actions, it is suggested the software future versions incorporate features that can detect things like drop-offs, and the localization has been tightened to be more error-aware and allows an operator to verify the robot's map alignment whenever there's doubt. Aside from these, the pilot's HSE record was excellent no other incidents occurred during hundreds of robot missions and over 1,000 hours of operation on DP4. The robot did not harm any person or equipment on the platform.

Another new feature that was tested and proved to be very valuable was the recent integration of a 4K 20x zoom camera on the robot. It did not only allow better quality images but opens up the possibility to capture images of smaller and further located objects, capture several checkpoints at the same robot location even with a single camera shot. The following figure shows the difference between two images, one taken with a normal camera and the second with a zoom camera.



Figure 8—Image comparison between robot cameras: (left) normal camera, zoom at 0%; (right) zoom at 30%

Remote Operations and Team Integration

The pilot found that a dedicated supervisor was indeed required to keep the robot productive and safe; the technology was not at a stage where it could just be left alone without anybody watching for an entire shift. Several training sessions before and during commissioning enabled the transfer of responsibility. The acceptance of the technology was crucial, because the robot often needed help that only on-site personnel could provide. The supervisor role became more routine as time went on and the system stabilized. Several autonomous missions were smoothly executed during night shift, data quality was found very good due to platform illumination, better contrast and visibility showing the potential for robotics activities around the clock.

The robot's activities were carried out under the platform's work permit system, ensuring no conflicts with other operations. It required daily meetings to plan the robot's schedule in coherence with other activities. During part of the pilot, a drilling rig was on DP4, meaning extra personnel and equipment on the platform requiring careful coactivity management, some robot missions were suspended. This approach paid off as even with high coactivity of humans and robot, there were no safety incidents or interferences.

Although the goal was minimal site intervention, the system required preventive, corrective maintenance and support. Things like replacing fixing a loose cable, or cleaning sensors still had to be done by humans on the platform. The O&M team based on Halul and the occasional visit of the robot supervisor handled these. In total, there were about 75 on-site interventions during the pilot execution period. The following graph shows a detailed breakdown of interventions:

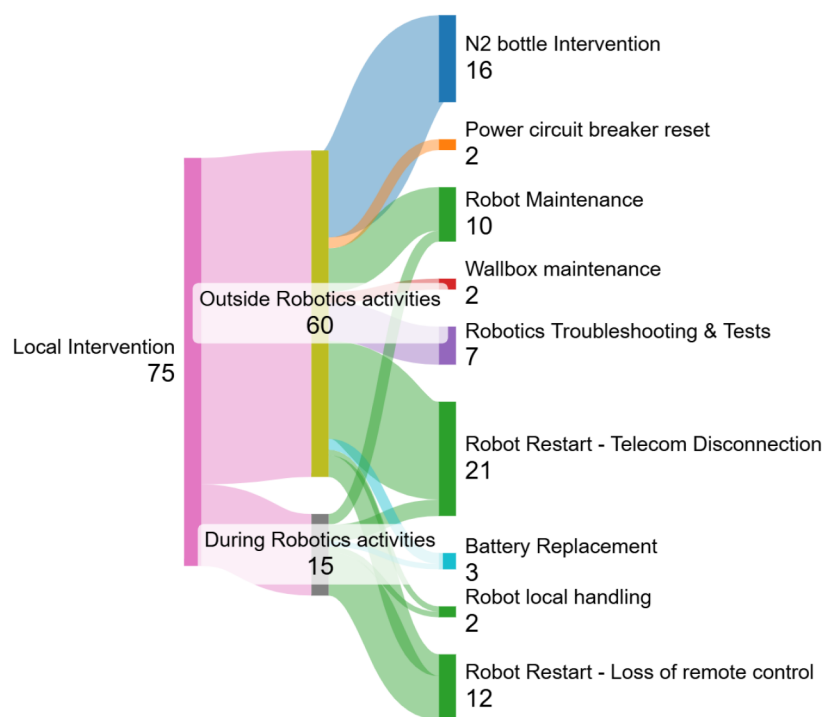


Figure 9—Local interventions distribution diagram

One out of five local interventions occurred outside robotics activities (no mission in progress) and as it can be seen, were mainly related to the robotics ancillary systems: telecommunications and nitrogen supply. For example, nitrogen consumption was higher than expected requiring replacing bottles more frequently. During robotics activities, the robot required sometimes to be restarted remotely to regain control of the system.

Remote expert support throughout the pilot was also provided. If something went wrong that the local team could not fix, specialists from the headquarters robotics team or robot constructor could log in to the systems (via the cloud platform or through a support laptop) to diagnose issues.

IT and Cybersecurity Infrastructure

The IT infrastructure was a hybrid solution: local control for real-time responsiveness and safety, coupled with cloud analytics for scalability and advanced data processing. This architecture not only met the technical needs of the pilot but also fit within corporate IT/OT (Information Technology / Operational Technology) governance, demonstrating that innovative robotic systems can be deployed respecting cybersecurity principles.

The IT team had to troubleshoot and adjust firewall settings as the pilot progressed to ensure all components talked as intended. Future deployments need thorough end-to-end network testing well ahead of time to avoid adjustment on the fly. Communication reliability was generally good. The wireless network on DP4 gave the robot coverage on most areas, though one blind spot was noted on the far southwest corner of the platform, opposite the Wi-Fi antennas.

Conclusions

The pilot met its key goals, validating the viability of the Normally Unattended Facility model in a real offshore setting. The autonomous robot successfully performed daily operator inspections (up to 500 checks per day) under remote supervision demonstrating the potential to reduce personnel exposure in remote offshore locations.

The robotic system proved it can inspect and monitor an unmanned installation for at least a week continuously and confirms that week-long (and longer) unattended operations are technically feasible with robotics. Notably, loss of remote control of the robot accounted for less than 3% of mission aborts, which validates the suitability of the IT and telecommunication setup most of the time. It marks a significant step toward making NUF a practical reality.

Quantitative results show significant improvements in reliability and efficiency. The pilot's MTBI improved 2.5 times compared to the previous pilot on a staffed facility. The robot's overall reliability was 27% higher than earlier trials. Mission success rate (tasks completed as planned) was 87%, with fewer than 3% of missions failing, evidence of robust remote connectivity.

The pilot underscored that a strong IT and communications infrastructure is as vital as the robot itself. A resilient wireless network and a well-architected backend (RSS on-site and RAS in the cloud) allowed seamless remote control and data flow. The system operated within TotalEnergies' stringent cybersecurity framework, segregating industrial control (IIS) and enterprise (EIS) networks without security incidents. The value also lied in its ability to leverage the acquired data for other business information systems, as it was the case during the pilot where two specific workflows were developed and successfully tested, one to transmit, store and visualize local operational data in the corporate, real-time software database and the second to generate a visual twin of the DP4 platform using images taken by the robot during the trials. In practice, the chosen IT architecture proved scalable and adaptable even under the field's network constraints. However, the pilot also highlighted the architecture will become more complex as data sharing increases and the need for careful planning of network aspects.

The integration of a more powerful camera, with a 20x zoom and higher resolution 4K, allowed reading and inspecting more elements that were not reachable before, and provides the opportunity to inspect several elements simultaneously, saving robot displacement time and increasing system productivity.

By combining these elements: a reliable robot, suitable sensors, artificial intelligence, a robust IT infrastructure, and trained remote operators, the project demonstrated concrete benefits. Safety can be improved, with thousands of inspection tasks completed without exposing staff to offshore hazards.

Operational continuity was enhanced: the platform remained under watch even during non-sailing days when crews normally couldn't visit. The data collected (nearly 10,000 asset measures) provided insights to O&M and support teams onshore, and critical anomalies (if they arose) would be caught in real-time.

The O&M team's feedback highlighted both the benefits of the robotic system and areas for improvement. They confirmed that the robot effectively covered routine Level 1 maintenance tasks and found its automated inspection reports valuable, suggesting that these reports be retained, refined, and made format customizable. The team is keen to expand the robot's capabilities by introducing additional use cases beyond the pilot's scope (such as fugitive gas leak detection or performing certain post-shutdown operations) and by eventually employing a manipulator robot in the future to handle more complex tasks like operating valves or switches. They also recommended designing the robot for greater flexibility in the field, for example, by making its sensors and tools easily swappable, and advised that alternative and simpler technologies (like fixed CCTV or IoT sensors) might suffice in place of the robot for efficiency. Importantly, the O&M team stressed that the overall system reliability (especially the docking station) needs to improve further, as too many on-site interventions were required during the trial; despite the positive trend observed, much higher reliability will be necessary for successful permanent deployment.

Despite the deployment costs of the robotics systems remain high compared to the amount of work completed on site, there are also early indications of cost and emission benefits, fewer boat trips translate to lower logistics costs and reduced CO₂ emissions. In short, the pilot's success shows that autonomous robotics offer the potential to make offshore operations safer and more efficient.

Building on these positive outcomes, it is recommended to expand and broaden what robots can do, improve machine learning models so a higher fraction of collected data is automatically interpreted to reduce burden on humans and unlock more value.

Finally, the robotic long-term trial showed that even if remote execution and supervision of inspection tasks are becoming a more tangible reality for unmanned installations, the operation is hardly hands-off; it requires a virtual team working in concert across locations to manage it; however it does decouple site operations from human presence on-site which validates the feasibility of the remote operations philosophy in the oil gas industry.

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- PERL, TotalEnergies' Research Centre in Lacq where TotalEnergies' robotics experimentation site is located.

Nomenclature

ADIPEC	Abu Dhabi International Petroleum Exhibition and Conference
AI	Artificial Intelligence
ATEX	ATmosphere EXplosibles (French: Explosive Atmospheres) equivalent to IECEx
CCR	Central Control Room
DMZ	DeMilitarized Zone
EIS	Enterprise Information System
FMS	Fleet Management System
FPSO	Floating Production Storage and Offloading (ship)
HMI	Human Machine Interface

IOGP	International Association of Oil & Gas Producers
IP	Ingress Protection
IP	Internet Protocol Security
ISS	Industrial Information System
IT	Information Technology
OT	Operational Technology
KPI	Key Parameter Indicator
LiDAR	Light Detection and Ranging
ML	Machine Learning
MTBI	Mean Time Between Intervention
NUF	Normally Unmanned Facility
O&G	Oil & Gas
O&M	Operation & Maintenance
RAS	Remote Analytics System
R&D	Research & Development
RSS	Robot Supervision System
SP1	Security Profile 1
SD-WAN	Software-Defined Wide Arranged Network
TADI	TotalEnergies Anomaly Detection Initiatives, TotalEnergies robotics lab
Wi-Fi	Wireless Fidelity

References

- Normally Unattended White Paper, IOGP - 2021 Normally Unattended Facilities white paper* | IOGP Publications library
- Munoz, JM and al. 2024. *First Integrated Robotics Pilot Offshore West Africa*. Paper SPE-222092-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 03-06 November
- Munoz, JM and al. 2023. *Ground Robotics: A Comprehensive Journey Towards Remote Operations*. Paper SPE-216285-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 02-05 October
- Bartoli, E, and al. 2022. *Implementation of Autonomous Ground Robots on Operational Sites*. Paper SPE-211242-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 31/10-03 November
- Audouin, G, and al. 2022. *First Proof of Concept for an Integrated Robot Supervision System*. Paper SPE-211039-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 31/10-03 November
- Munoz, JM, and al. 2022. *Generating Value From Inspection Ground Robots On Operational Sites*. Paper OTC-32112-MS presented at the Offshore Technical Conference (OTC), Houston, 02-05 May.
- Rossi, T, and al. 2022. *Unattended Floating Production Unit (FPU) and Attended Floating Storage Offloading (FSO)*. Paper OTC-31869-MS presented at the Offshore Technical Conference (OTC), Houston, 02-05 May.
- Mohd Rizal Mengan, Saiful Azuan Abdul Aziz, Nadirah Khairul Anuar, *Grant Veroba - Petronas, Jean-Michel Munoz - TotalEnergies*, 2022. OTC 31648 MS: Expanding Acceptance of Normally Unattended Facilities (NUF) - A Collaborative effort within IOGP, Offshore Technology Conference Asia, 22-25 March
- Munoz, JM, and al. 2020. *Operation Room: A Key Component for Upscaling Robotic Solutions on Site*. Paper SPE-202667-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 09-12 November.
- Munoz, JM, and al. 2019. *Ground Robotics Enabler For Normally Unattended Installations*. Paper SPE-197620-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 11-14 November
- Munoz, JM, and al. 2019. *Normally Unattended Installations, The New Frontier for Cost Reduction*. Paper SPE-197513-MS presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, 11-14 November.